

Local or remote?

How the subnet mask decides

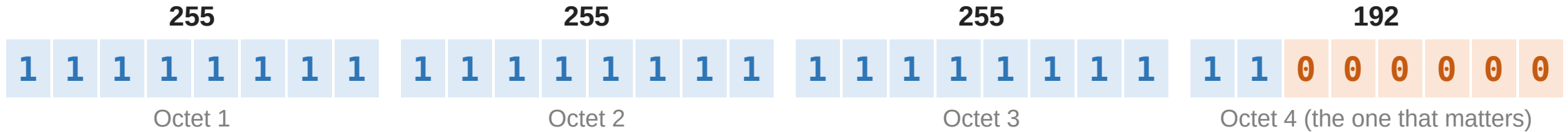
Why **192.168.10.37/26** and **192.168.10.70** are on **different subnets**, even though they look like neighbours.

192.168.10.37/26	→ 192.168.10.70	192.168.10.0 ≠ 192.168.10.64	Remote
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Contents: 1. What /26 means · 2. Blocks of 64 · 3. The AND proof · 4. The quick method · 5. Changing the mask · 6. What the host does · 7. Check your understanding

What does /26 mean?

/26 means the first **26 bits** identify the **network** and the remaining **6 bits** identify the **host** ($32 - 26 = 6$).



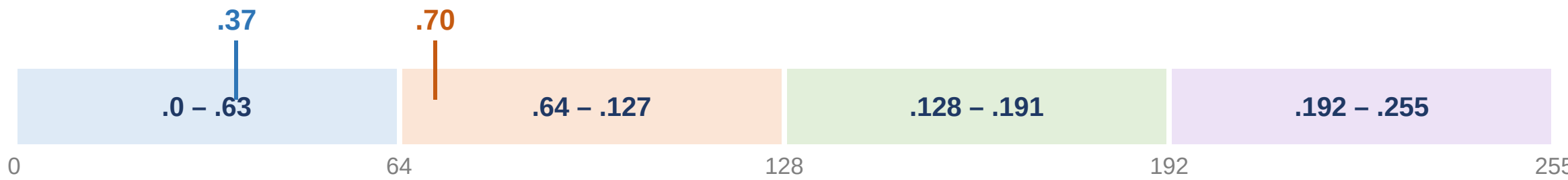
■ Network bits (mask = 1) ■ Host bits (mask = 0)

So the mask is 255.255.255.192

- The first three octets are all network bits: every address here starts with **192.168.10**.
- In the **fourth octet**, only the first **2 bits** (worth 128 and 64) are network bits. The last 6 bits are host bits.
- Those 2 network bits in the fourth octet are what separate .37 from .70.

6 host bits = blocks of 64 addresses

$2^6 = 64$ addresses per subnet. Shortcut: **block size = 256 – 192 = 64**. So 192.168.10.0 is cut into four subnets:



Subnet	Network address	Usable hosts	Broadcast	Contains
1	192.168.10.0	.1 – .62	.63	.37 is here
2	192.168.10.64	.65 – .126	.127	.70 is here
3	192.168.10.128	.129 – .190	.191	
4	192.168.10.192	.193 – .254	.255	

The proof: AND each address with the mask

A host finds a network address by **ANDing** the address with the mask, bit by bit: **1 AND 1 = 1**, anything else = 0. Only the fourth octet needs checking, because the first three are identical.

Source host: .37

37 0 0 1 0 0 1 0 1

192 1 1 0 0 0 0 0 0

AND 0 0 0 0 0 0 0 0 = 0

Network: 192.168.10.0

Destination: .70

70 0 1 0 0 0 1 1 0

192 1 1 0 0 0 0 0 0

AND 0 1 0 0 0 0 0 0 = 64

Network: 192.168.10.64

The whole difference is one bit, the one worth 64 (highlighted). The mask says that bit is a network bit, and .37 has a 0 there while .70 has a 1.

192.168.10.0 $\neq$ 192.168.10.64, so the destination is **REMOTE**.

The quick method (no binary needed)

Divide the last octet by the **block size**, keep the **whole number**, and multiply back:

Address	÷ 64	Whole number	× 64 = network
.37	37 ÷ 64 = 0.58	0	192.168.10.0
.70	70 ÷ 64 = 1.09	1	192.168.10.64

Rule: if the whole numbers differ, the addresses are on different subnets. 0 ≠ 1, so they are **remote**.

Block sizes to memorise (fourth octet)

Prefix	/24	/25	/26	/27	/28	/29	/30
Mask	.0	.128	.192	.224	.240	.248	.252
Block size	256	128	64	32	16	8	4
Usable hosts	254	126	62	30	14	6	2

Same two addresses, different masks

The addresses never change. **Only the mask changes**, and the answer flips from local to remote:

Prefix	Mask	Block size	Network of .37	Network of .70	Result
/24	255.255.255.0	256	.0	.0	Local
/25	255.255.255.128	128	.0	.0	Local
/26	255.255.255.192	64	.0	.64	Remote
/27	255.255.255.224	32	.32	.64	Remote

Key message: you cannot tell whether two addresses are neighbours by looking at the numbers. **/26 is the first prefix at which .37 and .70 separate.** Always apply the mask.

Trap to warn learners about: numbers that are close together can still be remote. For example, **.63 and .64** differ by only 1, but under /26 they are in different subnets (.63 is subnet 1's broadcast address, .64 is subnet 2's network address).

Why it matters: what the host actually does

Host 192.168.10.37/26 wants to send to 192.168.10.70. Is my network ID the same as the destination's?

YES → Local

Send an **ARP** request for .70's own MAC address.
Deliver the frame **directly** through the switch.

NO → Remote (our case)

Do **not** ARP for .70.
Send the frame to the **default gateway's** MAC address and let the **router** forward it.

Consequences to point out

- **No gateway configured means failure.** The ping fails even if both PCs are plugged into the **same switch**.
- **Same switch ≠ same subnet.** Two subnets always need a router (or layer-3 switch) between them.
- **Mismatched masks cause one-way failures.** If .70 were wrongly set to /24, it would treat .37 as local while .37 treats .70 as remote. Traffic works one way and fails the other. (Group 1, Task 4 demonstrates this in Packet Tracer.)

Check your understanding: local or remote?

Ask the class to work out each network ID before revealing the answers on the next page.

- 1 192.168.10.100/26 → 192.168.10.120
- 2 192.168.10.62/26 → 192.168.10.65
- 3 192.168.10.130/26 → 192.168.10.200
- 4 192.168.10.70/27 → 192.168.10.95
- 5 192.168.10.37/25 → 192.168.10.70
- 6 For 192.168.10.37/26: what are the network address, broadcast address and usable range?

Answers

#	Block size	Source network	Destination network	Result
1	64	$100 \div 64 \rightarrow 1 \rightarrow .64$	$120 \div 64 \rightarrow 1 \rightarrow .64$	Local
2	64	$62 \div 64 \rightarrow 0 \rightarrow .0$	$65 \div 64 \rightarrow 1 \rightarrow .64$	Remote (only 3 apart!)
3	64	$130 \div 64 \rightarrow 2 \rightarrow .128$	$200 \div 64 \rightarrow 3 \rightarrow .192$	Remote
4	32	$70 \div 32 \rightarrow 2 \rightarrow .64$	$95 \div 32 \rightarrow 2 \rightarrow .64$	Local
5	128	$37 \div 128 \rightarrow 0 \rightarrow .0$	$70 \div 128 \rightarrow 0 \rightarrow .0$	Local (the mask changed, not the addresses)

Question 6: $192.168.10.37/26 \rightarrow$ network **192.168.10.0**, broadcast **192.168.10.63**, usable **192.168.10.1 – 192.168.10.62** (62 hosts). The gateway is usually the first usable address, 192.168.10.1.

Summary for the class: apply the mask $\rightarrow$ compare network IDs $\rightarrow$ same means local (ARP directly), different means remote (send to the gateway).